L1-2: Homogeneous Coordinates

- E.g. for Pinhole Camera
  - Line- preserving
  - Not length-preserving (size inverse proportional to distance)
  - Not angle-preserving
- System of coordinates used in projective geometry
- Formulas involving H.C.: often **simpler** than Cartesian world
- Points at <u>infinity represented</u> using finite coordinates
  - Vanishing point: Point at infinity
- Single Matrix can represent affine and projective transfomations

Homogeneous Coordinates

Definition

- Representation $\mathbf{x}$ of geometric object is **homogeneous** if $\mathbf{x}$ and $\lambda\mathbf{x}$ represent same object for $\lambda \neq 0$

$$\underset{\text{homogeneous}}{\mathbf{x} = \lambda\mathbf{x}} \qquad \underset{\text{Euclidian}}{\mathbf{x} \neq \lambda\mathbf{x}}$$

- For Homogeneous Coordinate of point x in the plane $\mathbb{R}^2$ is a 3D vector

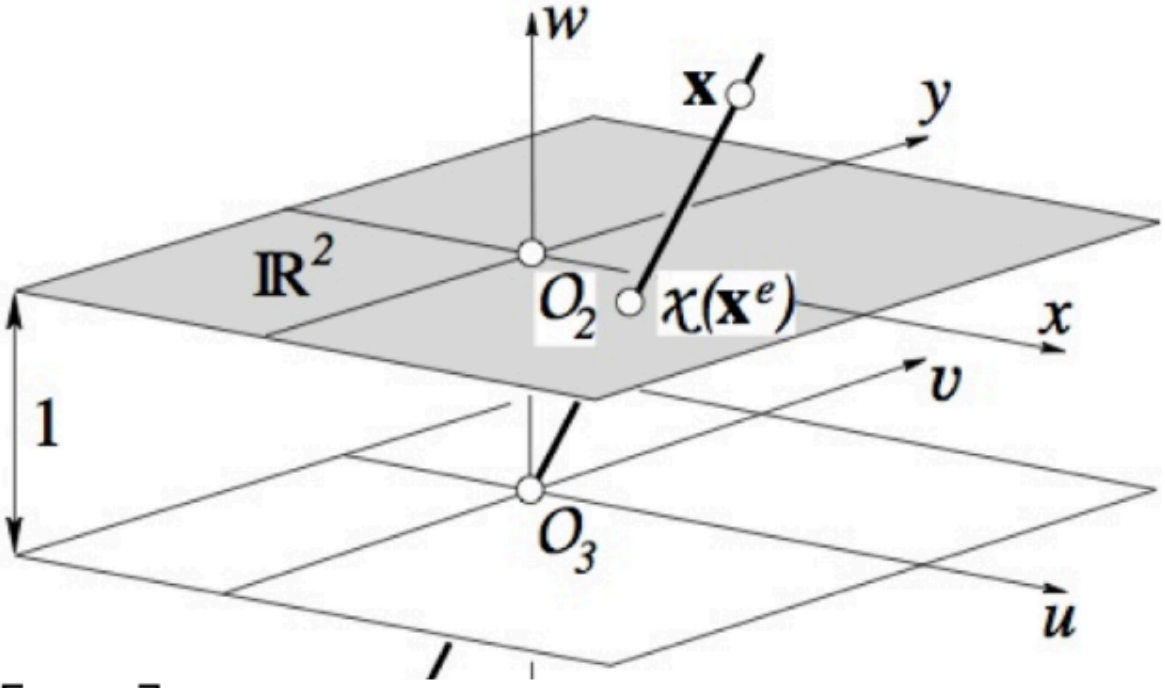
$$x : \mathbf{x} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \text{ with } |x|^2 = u^2 + v^2 + w^2 \neq 0$$

- Corresponds to Euclidian coordinates

$$x : \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u/w \\ v/w \end{bmatrix} \text{ with } w \neq 0$$

- H.C. use $n+1$ dimensional vector
  - for n-dimensional euclidian point
  - set dimension $n+1$ to 1

From Homogeneous to Euclidian Coordinates



$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} u/w \\ v/w \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} u/w \\ v/w \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Image courtesy

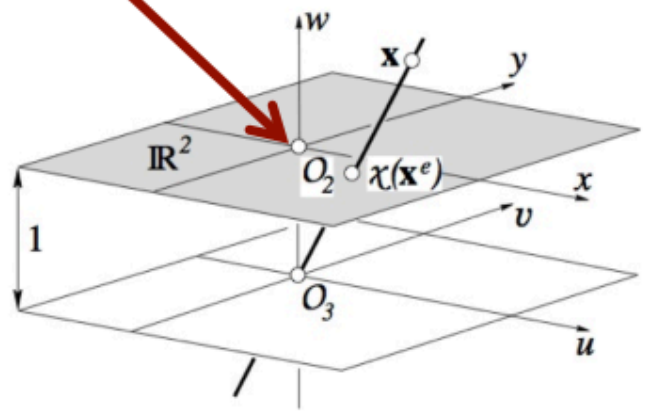
Analoguous for points in 3D Euclidian Space $\mathbb{R}^3$

Example

$$\underset{Euclidian}{\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix}} \quad \Rightarrow \quad \boldsymbol{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 2 \underset{homogeneous}{\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}} = \begin{bmatrix} 2x \\ 2y \\ 2 \end{bmatrix} = \begin{bmatrix} wx \\ wy \\ w \end{bmatrix}$$

Origin of the Euclidian Coordinate System in H.C.

$$\mathbf{O}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad \mathbf{O}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



Projective Plan

- Projective Plane $\mathbb{P}^2(\mathbb{R})$ or $\mathbb{P}^2$ contains
  - All points of the Euclidian Plane
  - All points at infinity ($\mathbf{x} = [x, y, 0]^T$)
  - Except $[0, 0, 0]^T$

Transformations

Projective Transformation is invertible linear mapping

$$\underbrace{\underline{\mathbf{X}'}}_{\text{homogeneous vector}} = \underbrace{\underline{\mathbf{H}}}_{\text{homogeneous matrix}} \underbrace{\underline{\mathbf{X}}}_{\text{homogeneous vector}}$$

- Every 1-to-1, straigth-line preserving mapping of a projective space $\mathbb{P}^n$ onto itself is a homography (projectibity) for $2 \leq n < \infty$
- Implies: All 1-to-1, straight-line preserving transformations are linear, when using projective coordinates

Translation

- 3 parameters (3 translations)

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathsf{H} = \lambda \begin{bmatrix} I & \underline{\boldsymbol{t}} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$$

$$\boldsymbol{t} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

homogeneous property

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Rotation

- 3 parameters

rotation matrix

$$\mathsf{H} = \lambda \begin{bmatrix} \underline{R} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$$

$$\mathsf{H} = \lambda \begin{bmatrix} \underline{R} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$$

- Rigid body transformation: 6 params
  (3 translation + 3 rotation)

$$\mathsf{H} = \lambda \begin{bmatrix} \underline{R} & \underline{\boldsymbol{t}} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$$

- Similarity transformation: 7 params
  (3 trans + 3 rot + 1 scale)

$$\mathsf{H} = \lambda \begin{bmatrix} \underline{mR} & \boldsymbol{t} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}$$

(angle-preserving)

- Affine transformation: 12 parameters
  (3 trans + 3 rot + 3 scale + 3 sheer)

$$\mathsf{H} = \lambda \begin{bmatrix} \underline{A} & \boldsymbol{t} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}$$

(not angle-preserving but parallel lines remain parallel)

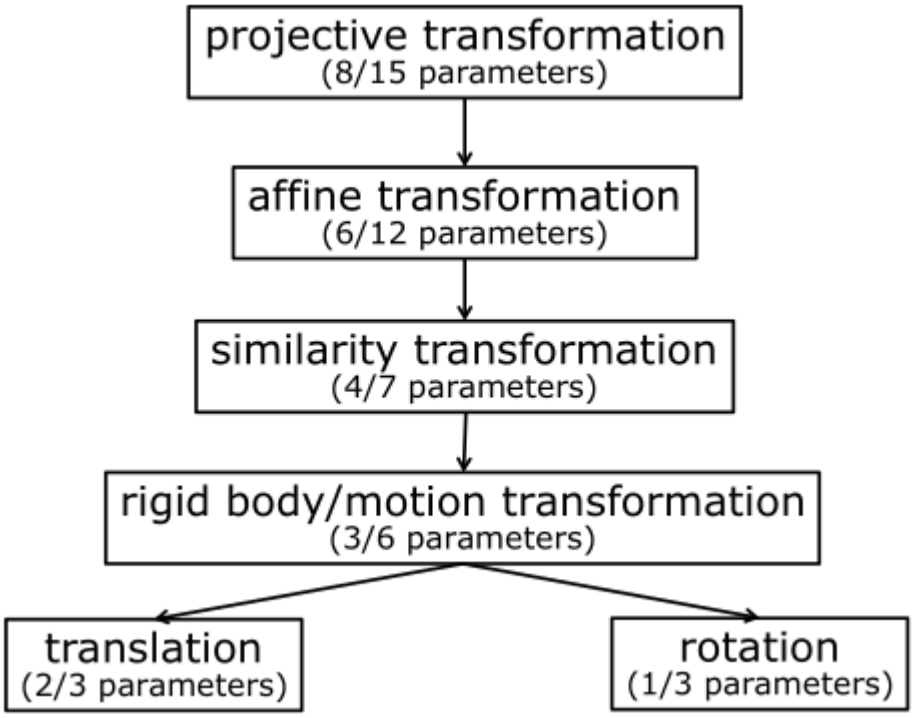
- Projective transformation: 15 params.

$$\mathsf{H} = \lambda \begin{bmatrix} A & \boldsymbol{t} \\ \underline{\boldsymbol{a}}^\mathsf{T} & 1 \end{bmatrix}$$

affine transformation + 3 parameters

- These 3 parameters are the projective part and they are the reason that **parallel lines may not stay parallel**

Transformation Hierarchy



Chaining and Inverting Transformations

- Chaining transformations via matrix products (not commutative)

$$\begin{aligned} X' &= H_1 H_2 X \\ &\neq H_2 H_1 X \end{aligned}$$

- Inverting a transformation

$$\begin{aligned} X' &= HX \\ X &= H^{-1} X' \end{aligned}$$

Representation of Lines

- Hesse normal form

$$x\cos\phi + y\sin\phi - d = 0 \quad \Rightarrow \quad (\cos\phi)x + (\sin\phi)y - d = 0$$

- Intercept form

$$\frac{x}{x_0} + \frac{y}{y_0} - 1 = 0 \quad \Rightarrow \quad \left(\frac{1}{x_0}\right)x + \left(\frac{1}{y_0}\right)y - 1 = 0$$

- Standard form

$$ax + by + c = 0 \quad \Rightarrow \quad ax + by + c = 0$$

All form linear equations that are equal to zero

Line Equation can be Expressed by the Dot-Product

$$\underset{\text{standard}}{\mathbf{l}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \qquad \underset{\text{Hesse}}{\mathbf{l}} = \begin{bmatrix} \cos\phi \\ \sin\phi \\ -d \end{bmatrix}$$

$$\underset{\text{intercept}}{\mathbf{l}} = \begin{bmatrix} \frac{1}{x_0} \\ \frac{1}{y_0} \\ -1 \end{bmatrix} \qquad \underset{\text{point}}{\mathbf{x}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{x} \cdot \mathbf{l} = 0$$

Definition

- Homogeneous Coordinates of a line l in the plane is 3-D Vector:

$$l : \mathbf{l} = \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} \text{ with } |\mathbf{l}|^2 = l_1^2 + l_1^2 + l_3^2 \neq 0$$

- Corresponds to the Euclidian representation

$$l_1 x + l_2 y + l_3 = 0$$

Point on Line

- Point

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Lies on line

$$\mathbf{l} = \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix}$$

- IF $\mathbf{x} \cdot \mathbf{l} = 0$

Intersection Lines

- Intersection $x = l \cap m$
- Find Point $\mathbf{x} = [x, y]^T$

$$\begin{bmatrix} \mathbf{l} \cdot \mathbf{x} \\ \mathbf{m} \cdot \mathbf{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} l_1 & l_2 \\ m_1 & m_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -l_3 \\ m_3 \end{bmatrix}$$

Or, using Cramer Rule

$$\mathbf{x} = \mathbf{l} \times \mathbf{m}$$

Line between 2 Points

- $l = x \wedge y : \quad \mathbf{l} = \mathbf{x} \times \mathbf{y}$

Points and Lines at Infinity

- Points at Infinity
  - $x_\infty : \mathbf{x}_\infty = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}$
  - Maintain direction to infinitvely distant point
  - Great for bearing-only sensors (camera)
- Intersection at Infinity
  - All lines l with $l \cdot x_\infty = 0$ pass through x_∞
  - <u>All parallel lines meet at one point at infinity</u>

- This means $[u, v] \cdot [\cos\phi, \sin\phi] = 0$
- This hold for any line $\mathbf{m} = [\cos\phi, \sin\phi, *]^T$ i.e. for any line that is parallel to ℓ
- This can also be seen by

$$\mathbf{l} \times \mathbf{m} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \times \begin{bmatrix} a \\ b \\ d \end{bmatrix} = \begin{bmatrix} bd - bc \\ ac - ad \\ ab - ab \end{bmatrix} = \begin{bmatrix} bd - bc \\ ac - ad \\ 0 \end{bmatrix}$$

- Infinitively Distant Objects:
- Infinitively distant Point:
  - $x_\infty : \quad \mathbf{x}_\infty = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}$
- Infinitively distant line (ideal line)
  - Can be seen as the **horizon**
  - $l_\infty : \quad \mathbf{l}_\infty = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
- All points at infinity lie on this line
  - $x_\infty \mathbf{l}_\infty = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$

Analogous for 3 Objects

- 3D point

$$\mathbf{X} = \begin{bmatrix} U \\ V \\ W \\ T \end{bmatrix} = \begin{bmatrix} U/T \\ V/T \\ W/T \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} U/T \\ V/T \\ W/T \end{bmatrix}$$

- Plane

$$\mathbf{A} = \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix}$$

Point on a Plane

- Via the scalar product, we can again test if a point lies on a plane

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{A}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{A} = 0$$

- which is based on

$$AX + BY + CZ + D = 0 \quad \text{or} \quad N \cdot X - S = 0$$

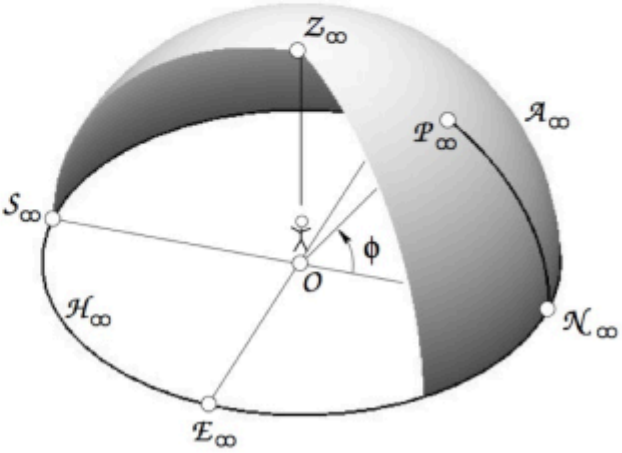
3D Objects at Infinity

- 3D point

$$\mathbf{P}_\infty = \begin{bmatrix} U \\ V \\ W \\ 0 \end{bmatrix}$$

- Plane

$$\mathbf{A}_\infty = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



Summary

- Homogeneous coordinates are an alternative representation for geometric objects
- Can simplify mathematical expressions
- Can model points at infinity
- Easy chaning and inversion of transformations
- Uses extra dimension
- Equivalence up to scale